

Abstract

Abstract I

Abstract II

Commutative diagrams are cognitively privileged representations because a single diagram simultaneously encodes three things a cognitive agent needs at once: the algebraic structure of a relational situation, the distributional semantics by which language reports on that situation, and the inference process by which belief is updated when new evidence arrives; the traditional linguistic category of case — who did what to whom — is the natural fulcrum on which all three layers turn. We review formalized linguistic case systems as categories whose objects are case roles and whose morphisms are grammatical relations, and we capture every cross-linguistic alignment type (nominative-accusative, ergative-absolutive, tripartite, active-stative, fluid-S) as a structure-preserving functor between case categories. Sentences and discourses become compact-closed string diagrams via DisCoCat and DisCoCirc, computable as executable witnesses of grammatical and cognitive composition; enriching the case category with $[0,1]$ -weighted hom-values supplies quantitative measures and an explicit link to distributional proximity; a topos-theoretic bridge then ties the typological type-logical